

Phones to make.

Negative binomial distribution

A r.v. X is said to follow a negative binomial distribution if its probability mass function is given by

$$p(n) = P(X=n) = \binom{n+r-1}{r-1} p^r q^n$$

Remarks

$$n = 0, 1, 2, \dots$$
$$= -r \ln p (-q)^n, \quad n = 0, 1, 2, \dots$$

$$\text{mean } \frac{rq}{p}, \quad \text{var} = \frac{rq}{p^2}$$